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Task 1

- Chosen sentences:
    - Large language models are transforming natural language processing by enabling machines to generate, summarize, and translate text at human-like levels.
    - However, these systems require large datasets, extensive computation, and careful evaluation to ensure accuracy, fairness, and reliability in real-world applications.
- This summary is grammatically fine and meaningful, but it’s really just a teaser. it’s not the full story. It stuck to the sentence limit but cut out crucial information to do so (the failures of standard metrics and prompt engineering).

Task 2

Extractive vs. Abstractive Summaries:

The abstractive summary is better because it covered all the main points (LLMs are good, but evaluation is tricky, and prompt engineering is a mess). The extractive one missed the part about prompts and biased data.

The abstractive summary is easier to read. It sounds like a person wrote it, making a nice, short paragraph that makes sense.

The extractive summary is safer. It just copies the original words, so it can't accidentally make up facts (hallucinate) or mess up the author's meaning.

I would trust the abstractive one for a quick idea of the whole article, but I'd trust the extractive one if I needed to make sure the quotes were exactly right.

Task 3

score(1-5)	Extraction			Abstraction		
	F	C	Faith	F	C	Faith
Sample 1	2	5	5	5	3	5
Sample 2	2	5	5	4	4	5
Sample 3	2	5	5	4	3	5
Sample 4	4	3	5	5	3	5
Sample 5	4	2	5	5	2	5
Sample 6	4	2	5	5	4	5

Faithfulness
- True to the original text (Contains only info from the source)
- No distortions

Comparison

1. Handling Single Line Long Sentence (Sample 1,2,3):

- **Extractive (TextRank) failed** because it relies on punctuation (periods) to identify sentences. It treated the entire messy input as one block and returned the whole text.
- **Abstractive (BART) excelled** by understanding the meaning and rewriting the "run-on" text into clear, properly punctuated sentences.

2. Handling Normal Multiple Technical Sentences (Sample 4,5,6):

- **Extractive (TextRank) was accurate but disjointed.** It successfully pulled key, high-keyword-density sentences, but these selected sentences often lacked a logical flow or missed the document's main setup, focusing only on conclusions or details.
- **Abstractive (BART) was coherent but sometimes incomplete.** It produced smooth, human-like summaries but occasionally cut off the end of the document due to length constraints, missing the conclusion.

3. Key Trade-offs:

- **Faithfulness (Accuracy): Extractive is the winner** as it only uses text from the source and never "hallucinates." Abstractive is good but risks losing subtle nuance by interpreting technical terms.
- **Fluency (Readability): Abstractive is the clear winner.** It creates a coherent narrative, fixing grammar and flow. Extractive often results in a choppy, illogical paragraph.